

AN INTERNSHIP REPORT ON

***Solar-Powered IoT System for Real-Time Water Quality
Monitoring in Aquaculture***

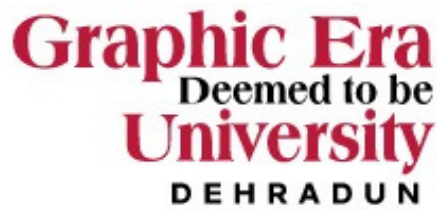
***BACHELOR OF SCIENCE
IN
ELECTRICAL AND ELECTRONICS
ENGINEERING***

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CANDIDATE'S DECLARATION

I hereby certify that the work is being presented in the Project Report entitled "Solar-Powered IoT System for Real-Time Water Quality Monitoring in Aquaculture" in partial fulfillment of the requirements for the award of the Degree of Bachelor of Science in Electrical and Electronics Engineering and submitted to the Department of Electronics and Communication Engineering of the Graphic Era (Deemed to be University), Dehradun is an authentic record of my work carried out during a period from July-2026 to August-2026 under the supervision of Dr. Gourav Verma and approved by Dr. Abhay Sharma, Professor, Department of Electronics and Communication Engineering, Graphic Era (Deemed to be University).

The matter presented in this dissertation has not been submitted by me for the award of any other degree of this or any other Institute/University.

Yunus Kunduz

This is to certify that the above statement made by the candidate is correct to the best of our knowledge.

Signature

Dr. Gourav Verma

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1. INTRODUCTION

1.1 Project Overview & Objectives

The rapid growth of the aquaculture industry requires continuous and precise monitoring of water parameters to ensure aquatic health and maximize yield. Traditional manual sampling methods are time-consuming, labor-intensive, and fail to provide real-time insights. As environmental and operational demands increase, there is a critical need for autonomous systems capable of continuous data acquisition and remote monitoring.

The primary objective of this internship project was to design and implement a solar-powered Internet of Things (IoT) node for real-time water quality monitoring. Instead of relying on conventional grid power, the proposed system operates autonomously using a solar panel and a battery management circuit, ensuring continuous operation even in remote environments.

The project also aimed to demonstrate how embedded systems and cloud technologies can be combined to solve real-world agricultural problems. Throughout the internship, the system was developed by following engineering principles such as hardware integration, power management design, microcontroller programming, and cloud platform configuration. [1]

Besides technical implementation, the internship provided an opportunity to gain practical experience in modern electronics and IoT methodologies. The completed system consists of an ESP32 microcontroller, an analog turbidity sensor, a temperature probe, a custom power management circuit (including an MT3608 boost converter), and a web-based dashboard on Thinger.io. Together, these components create a reliable, low-power architecture capable of real-time environmental monitoring. [2]

1.2 Problem Statement

Despite significant technological advancements in modern agriculture, many aquaculture facilities still heavily rely on manual water quality testing and traditional observation methods. This conventional approach only provides periodic data snapshots rather than continuous monitoring. Sudden fluctuations in critical water parameters—such as water temperature and turbidity—can occur rapidly due to sudden weather changes, biological waste buildup, or equipment failures. When these parameters fall out of the optimal range, it can lead to severe consequences, including dissolved oxygen depletion, increased susceptibility to diseases, and significant mortality rates in aquatic life. Manual sampling introduces delayed reaction times, meaning that by the time a problem is detected by facility operators, the biological damage is often irreversible.

Another major limitation of existing commercial monitoring systems is their dependency on a continuous electrical grid or the frequent replacement of non-rechargeable batteries. In many rural, off-grid, or offshore aquaculture environments, accessing a stable power infrastructure is either technically impractical or economically prohibitive. Systems relying solely on standard batteries require constant manual maintenance, which increases labor costs and defeats the primary purpose of an automated monitoring system. Therefore, for a monitoring system to be truly effective and scalable in these remote locations, it must be highly energy-efficient and self-sustaining. [3]

Furthermore, isolated measurement tools fail to provide historical data tracking and remote accessibility. Farm managers require the ability to monitor their facilities from any location and receive immediate alerts when water parameters reach critical thresholds. Without proper cloud integration, data remains localized, preventing comprehensive analysis and proactive decision-making.

To overcome these combined biological, logistical, and infrastructural challenges, this project introduces an autonomous, solar-powered IoT solution. By integrating efficient solar energy harvesting with low-power microcontrollers and wireless data transmission, the system continuously tracks water quality and relays data to a centralized cloud dashboard, completely eliminating the need for daily manual intervention and dedicated power grid connections. [1]

1.3 Scope of the Project

The scope of this internship project covers the end-to-end development of an autonomous IoT hardware node, beginning with component selection and circuit design, and concluding with cloud integration and field evaluation. The project was designed as a modular embedded system that demonstrates how renewable energy and IoT technologies can be combined to create a reliable, scalable monitoring solution for aquaculture.

The first phase focused on hardware architecture and power management. A custom PCB docking board was utilized to house an ESP32 microcontroller, integrated with a solar charge controller and an MT3608 boost converter. This power management layer was specifically designed to harvest energy from a solar panel to sustain a 4-cell battery pack, ensuring continuous off-grid operation.

The second phase involved sensor integration and data acquisition. A commercial analog turbidity sensor and a waterproof metal-tipped temperature probe were interfaced with the microcontroller. Proper voltage divider networks and Analog-to-Digital Converter (ADC) mapping techniques were implemented to accurately read environmental parameters and safely track the real-time battery voltage. [2]

Cloud integration and software development constituted another significant component of the project. The ESP32 was programmed to periodically sample sensor data and transmit it over a local Wi-Fi network. The backend integration was established using the Thingier.io platform, where a graphical dashboard was developed to visualize real-time water temperature, turbidity levels, and system battery status. Additionally, the architecture incorporated Sinric Pro to bridge the system with smart home ecosystems, enabling voice-controlled queries via Amazon Alexa. [3]

Finally, practical field testing was conducted to validate the functionality, energy efficiency, and reliability of the complete system. The hardware was exposed to variable weather conditions over a one-week period, during which battery voltage levels were manually tracked using a digital multimeter. The system's power harvesting capability in cloudy conditions, sensor data accuracy, and stable cloud communication were rigorously evaluated to ensure stable performance.

Overall, the project demonstrates the successful integration of embedded hardware, renewable energy management, and modern IoT cloud platforms into a practical, self-sustaining solution capable of addressing real-world challenges faced by contemporary aquaculture facilities. [2]

2. INSTITUTION PROFILE

2.1 Graphic Era (Deemed to be University)

The internship was conducted at Graphic Era (Deemed to be University), located in Dehradun, Uttarakhand, India. Established in 1996 as the Graphic Era Institute of Technology, the institution was granted the status of a "Deemed to be University" in 2008 by the University Grants Commission (UGC). Today, it is recognized as one of India's leading universities, particularly in engineering, technology, and research.

The university provides a modern learning environment equipped with advanced hardware laboratories, digital libraries, and research facilities. These resources support both academic education and practical R&D activities, allowing students and international interns to gain hands-on experience alongside theoretical knowledge.

Throughout this AIESEC international internship program, the university maintained a supportive academic atmosphere where faculty members provided valuable guidance. Working in this cross-cultural environment improved not only technical engineering skills but also global communication, teamwork, and professional development.

2.2 Department of Electronics and Communication Engineering (ECE)

The internship project was completed within the Department of Electronics and Communication Engineering (ECE), which focuses on embedded systems, Internet of Things (IoT), hardware design, telecommunications, and sustainable energy applications. The department offers modern laboratories equipped with the electronic instruments, microcontrollers, and measurement tools required for practical hardware development.

At the beginning of the internship, I was introduced to the ECE laboratory facilities and available development resources. Meetings with the academic supervisor, Dr. Gourav Verma, helped define the IoT project scope and provided guidance on the hardware integration practices that would be followed throughout the internship.

The department encouraged the use of practical engineering methodologies such as custom PCB design, power management analysis, and sensor calibration. Continuous discussions with the faculty helped improve the overall quality of the solar-powered node and provided solutions to physical implementation challenges encountered during the assembly and testing phases.

Overall, the ECE Department offered an excellent environment for learning and applying modern embedded systems and IoT techniques. Its research-oriented approach and experienced faculty members contributed significantly to the successful prototyping and completion of the internship project.

3. HARDWARE ARCHITECTURE AND POWER MANAGEMENT

3.1 High-Level Architecture

The developed system follows a modular hardware architecture to ensure autonomy, reliability, and efficient power management in remote aquaculture environments. The architecture is divided into three main layers: the Power Management Layer, the Data Acquisition Layer, and the Processing & Communication Layer. Each layer performs specific tasks while interacting with the others through custom-designed PCB routing.

The Power Management Layer is responsible for sustaining the autonomous operation of the node. It consists of a 4-cell battery pack that is continuously recharged by a solar panel. An onboard solar charge controller, coupled with an MT3608 boost converter module, regulates the incoming solar energy to safely charge the batteries and provide a stable operating voltage. [5]

The Data Acquisition Layer interacts directly with the physical environment. It incorporates a waterproof metal-tipped probe for temperature measurement and a commercially available analog turbidity sensor. Additionally, a voltage divider network is integrated to safely step down and read the battery voltage levels. [7]

The Processing & Communication Layer acts as the core of the system. An ESP32 microcontroller processes the analog and digital signals from the sensors, applies necessary calibration factors, and manages the Wi-Fi communication. It ensures that the acquired data is reliably transmitted to the cloud interfaces. This layered architecture separates hardware responsibilities, making the system easier to maintain, test, and deploy. [4]

3.2 Hardware Components & Technology Stack

Several modern electronic components and IoT frameworks were used during the development of the project to ensure high performance and reliability.

ESP32 Microcontroller was selected as the primary processing unit due to its integrated Wi-Fi capabilities, robust processing power, and efficient power management suitable for IoT applications.

Custom PCB Docking Board was utilized to securely mount the ESP32 and integrate the voltage divider network, ensuring reliable pin connections and overall system modularity. [4]

MT3608 Boost Converter & Solar Controller served as the core of the power management system. These modules safely regulate solar energy input and step up the voltage to power the microcontroller consistently, even under predominantly cloudy weather conditions. [5]

Environmental Sensors included a digital waterproof metal-tipped temperature probe and an analog turbidity sensor to accurately capture real-time water quality metrics without the risk of corrosion. [6,7]

Thingier.io was utilized as the primary IoT cloud platform. It enabled the rapid creation of an interactive web dashboard for real-time visualization of sensor data and battery status. [8]

Sinric Pro was integrated to bridge the ESP32 hardware with smart home ecosystems, enabling voice-controlled interactions and status queries via Amazon Alexa. [9]

4. SOFTWARE DESIGN AND CLOUD INTEGRATION

4.1 Embedded Software and Sensor Calibration

The software architecture of the ESP32 node was developed using the Arduino Integrated Development Environment (IDE). The core logic was written in C/C++, and iterative testing was conducted directly on the ESP32 hardware to ensure stable data acquisition and processing. The microcontroller was programmed to periodically sample analog data from the turbidity sensor and the battery voltage divider network. For water temperature acquisition, the system utilizes a digital communication protocol to interface with the metal-tipped probe, ensuring high-precision readings. Once the raw sensor data is acquired, the ESP32 processes these values locally, applying calibration algorithms to convert analog ADC (Analog-to-Digital Converter) readings into meaningful metrics, such as actual voltage levels and turbidity percentages. To maintain system stability, the software includes timing routines that manage the frequency of sensor sampling and prevent network congestion during Wi-Fi data transmissions. [10]

4.2 IoT Cloud Platform Integration (Thingier.io)

To enable remote monitoring, the processed data is transmitted via the ESP32's built-in Wi-Fi module to the Thingier.io IoT platform. A custom dashboard was developed on Thingier.io to provide real-time visualization of critical parameters. The dashboard is structured to display historical trends using time-series charts for Water Turbidity, Water Temperature, and Battery Voltage, alongside interactive gauge widgets for instant status checks. This cloud integration allows aquaculture operators and farm managers to access live environmental data seamlessly through web or mobile interfaces, eliminating the need for physical on-site inspections. [8]

4.3 Smart Home Integration via Sinric Pro

To further enhance the accessibility and user experience of the monitoring system, the software architecture incorporates Sinric Pro to bridge the ESP32 node with modern smart home ecosystems. This API integration enables voice-controlled interactions via Amazon Alexa. Farm operators can effortlessly query the current status of the aquaculture environment—such as asking for the real-time water temperature or checking if the battery level is sufficient—using natural voice commands. This hands-free dimension introduces a highly modern, efficient, and user-friendly approach to agricultural monitoring. [9]

5. FIELD IMPLEMENTATION AND TESTING

To validate the reliability and autonomy of the proposed IoT system, practical field tests were conducted over a one-week period. A critical phase of the testing focused on the efficiency of the solar-powered energy management system. The battery voltage levels were manually tracked and recorded day by day using a digital multimeter to verify the performance of the solar charge controller and the MT3608 boost module.

At the beginning of the observation period, the initial battery voltage was measured at 3.08V. Over the course of approximately one week, the system was exposed to predominantly cloudy weather conditions, including a brief period of rain. Despite the lack of direct and strong sunlight, the solar panel and power management circuit successfully harvested enough energy to charge the 4-cell battery pack. By the end of the observation period, the multimeter readings indicated a voltage increase to approximately 3.60V.

This steady rise in voltage under suboptimal, cloudy weather conditions strongly validates the high efficiency of the power hardware architecture. Furthermore, the real-time sensor data and battery status were consistently and successfully transmitted to the Thingier.io dashboard, confirming the stability of the ESP32 Wi-Fi communication and the overall system integration. [5]

6. CONCLUSION

The primary objective of this internship project was successfully achieved through the design, implementation, and testing of an autonomous, solar-powered IoT system for real-time water quality monitoring. By integrating an ESP32 microcontroller with precise environmental sensors and a robust power management circuit, the project demonstrated a practical and scalable solution for continuous monitoring in aquaculture facilities.

The hardware architecture proved its reliability and energy efficiency during field testing. The custom power management system successfully sustained the operation of the remote node and charged the battery pack even under predominantly cloudy weather conditions. This validated the system's capability for off-grid operation, significantly reducing the need for manual maintenance and reliance on conventional power grids.

Furthermore, the embedded software, developed using the Arduino IDE, established seamless cloud communication. The integration with Thingier.io provided a user-friendly dashboard for real-time data visualization, while the implementation of Sinric Pro introduced modern, voice-controlled accessibility via Amazon Alexa.

Overall, this project provided comprehensive practical experience in embedded systems, hardware design, power electronics, and modern IoT cloud platforms. The developed prototype stands as a highly effective tool that addresses real-world logistical challenges in modern agriculture, ensuring optimal water conditions through continuous, automated, and self-sustaining environmental monitoring. [8, 9, 10]

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